

## Week - 8 First Order Logic

Q. Create a knowledge base consisting of first order logic statement & prove the given query using forward reasoning

premises  $P \Rightarrow Q \rightarrow$  Conclusion

Rules

$$L \wedge M \Rightarrow P$$

$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

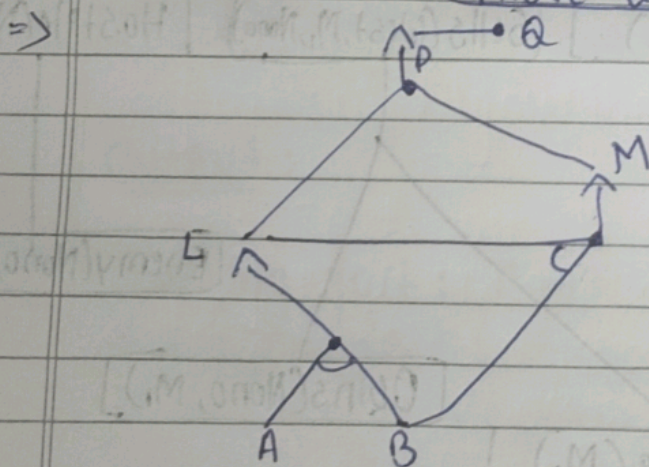
$$A \wedge B \Rightarrow L$$

facts

A

B

Prove Q



Q. The law says that it is a crime for an American to sell weapons to hostile nations. The Country Nono an enemy of America, has some missiles & all of its missiles were sold to it by Colonel West who is American. An enemy of America counts as 'hostile' Prove that "West is Criminal"

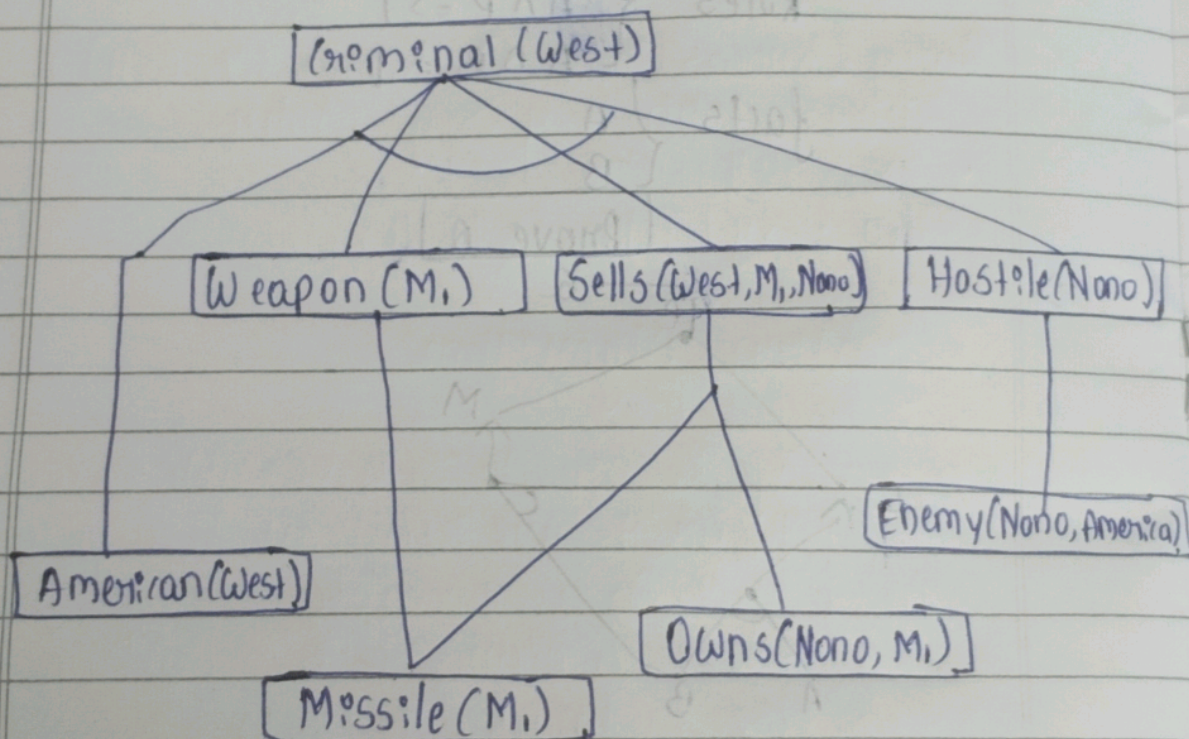
$\Rightarrow$  Rules

1)  $\forall x, y, z \text{ American}(x) \wedge \text{weapon}(y) \wedge$



$Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$

- Rules
- 1)  $\forall x Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$
  - 2)  $\forall x Enemy(x, America) \Rightarrow Hostile(x)$
  - 3)  $\forall x Missile(x) \Rightarrow Weapon(x)$
  - 4)  $American(West)$
  - 5)  $Enemy(Nono, America)$
  - 6)  $Owns(Nono, M1)$  and
  - 7)  $Missile(M1)$



### Algorithm

function FOL-FC-Ask( $KB, \alpha$ ) returns a substitution or false

inputs:  $KB$ , the knowledge base, a set of first-order definite clauses  
 $\alpha$ , the query, an atomic sentence  
 local variables: new, the new sentences inferred on each iteration



Repeat until new is empty

new  $\leftarrow \{\}$

for each rule in KB do

$(p' \wedge \dots \wedge p_n \Rightarrow q) \leftarrow \text{STANDARDIZE-VARIABLES}(\text{rule})$

for each  $\theta$  such that  $\text{SUBST}(\theta, p', \dots, p_n) = \text{SUBST}(\theta, p'_1 \wedge \dots \wedge p'_n)$

for some  $p'_1, \dots, p'_n$  in KB

$q' \leftarrow \text{SUBST}(\theta, q)$

if  $q'$  does not unify with some sentence already in KB or new then

add  $q'$  to new

$\theta \leftarrow \text{UNIFY}(q', \alpha)$

if  $\theta$  is not fail then return  $\phi$

add new to KB

return false.

Output:

Adding fact: American(West)

Adding fact: Enemy(Nono, America)

Adding fact: Missile(M1)

Adding fact: Owns(Nono, M1)

Inferred new fact: Hostile(Nono) from

$['\text{Enemy}(\text{Nono}, \text{America})'] \Rightarrow \text{Hostile}(\text{Nono})$

Inferred new fact: Weapon(M1) from  $['\text{Missile}(\text{M1})'] \Rightarrow \text{Weapon}(\text{M1})$

Inferred new fact: Sells(West, M1, Nono) from  $['\text{Missile}(\text{M1})', '\text{Owns}(\text{Nono}, \text{M1})'] \Rightarrow \text{Sells}(\text{West}, \text{M1}, \text{Nono})$

Inferred new fact: Criminal(West) from  $['\text{American}(\text{West})', '\text{Weapon}(\text{M1})', '\text{Sells}(\text{West}, \text{M1}, \text{Nono})', '\text{Hostile}(\text{Nono})'] \Rightarrow \text{Criminal}(\text{West})$

Goal Reached: West is criminal  
True.

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Adding fact: American(West)
Adding fact: Enemy(Nono, America)
Adding fact: Missile(M1)
Adding fact: Owns(Nono, M1)
Inferred new fact: Hostile(Nono) from ['Enemy(Nono, America)'] => Hostile(Nono)
Inferred new fact: Weapon(M1) from ['Missile(M1)'] => Weapon(M1)
Inferred new fact: Sells(West, M1, Nono) from ['Missile(M1)', 'Owns(Nono, M1)'] => Sells(West, M1, Nono)
Inferred new fact: Criminal(West) from ['American(West)', 'Weapon(M1)', 'Sells(West, M1, Nono)', 'Hostile(Nono)'] => Criminal(West)

Goal Reached: West is Criminal
True
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